Course Code | Course Name | Teaching Scheme

2345111 Software Engineering and Agile Practices

Theory Practical Tutorial Credits

3 0 0 3

Course Objectives:

1. Understand core concepts.

2. Apply knowledge practically.

Course Outcomes:

CO1: Demonstrate knowledge of fundamentals.

CO2: Analyze and design solutions.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Data Structures, Mathematics, Programming	00	
1	SDLC and Process Models	waterfall model, incremental model, spiral model, RAD model, COCOMO estimation, function point analysis, project planning		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
2	Requirements Engineering	use case diagrams, user stories, requirements elicitation, functional and non-functional requirements, prototyping, RMMM risk matrix		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
3	Agile Methodologies	scrum framework, sprint planning, daily standup, kanban board, retrospective, velocity tracking, burndown chart		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
4	Software Design	UML diagrams, class diagrams, sequence diagrams, design patterns, SOLID principles, cohesion and coupling		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
5	Software Testing	unit testing, integration testing, system testing, regression testing, black-box testing, white-box testing, equivalence partitioning, test-driven development		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
6	Maintenance and Quality	software maintenance types, refactoring, code review, SQA metrics, ISO standards, configuration management		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1

Total Hours: 48

Note: No questions from Self-Learning topics.

Text Books:

1. Web Development by J. Smith, Pearson, 2nd edition.

References:

1. MDN Web Docs. https://developer.mozilla.org

Assessment:

IAT-I + IAT-II: 40 Marks, End Sem: 60 Marks

Course Code | Course Name | Teaching Scheme

2345112 AI and ML

Theory Practical Tutorial Credits

3 0 0 3

Course Objectives:

1. Understand core concepts.
2. Apply knowledge practically.

Course Outcomes:

CO1: Demonstrate knowledge of fundamentals.

CO2: Analyze and design solutions.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Data Structures, Mathematics, Programming	00	
1	AI Foundations and Search	problem formulation, state space search, breadth first search, depth first search, A* algorithm, heuristic functions, adversarial search Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
2	Knowledge Representation	propositional logic, first-order logic, Bayesian networks, inference rules, resolution refutation Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
3	Supervised Learning	linear regression, logistic regression, decision trees, random forest, support vector machine, naive Bayes classifier, gradient descent Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
4	Unsupervised Learning	k-means clustering, hierarchical clustering, principal component analysis, dimensionality reduction, association rule mining, apriori algorithm Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
5	Model Evaluation	confusion matrix, precision and recall, F1 score, ROC curve, AUC score, cross-validation, bias-variance tradeoff Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
6	Neural Networks	perceptron model, multilayer perceptron, backpropagation, activation functions, convolutional neural networks, recurrent neural networks, transfer learning Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1

Total Hours: 48

Note: No questions from Self-Learning topics.

Text Books:

1. Web Development by J. Smith, Pearson, 2nd edition.

References:

1. MDN Web Docs. https://developer.mozilla.org

Assessment:

IAT-I + IAT-II: 40 Marks, End Sem: 60 Marks

Course Code | Course Name | Teaching Scheme

2345113 Web Technology

Theory Practical Tutorial Credits

3 0 0 3

Course Objectives:

1. Understand core concepts.

2. Apply knowledge practically.

Course Outcomes:

CO1: Demonstrate knowledge of fundamentals.

CO2: Analyze and design solutions.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Data Structures, Mathematics, Programming	00	
1	Web Fundamentals and JavaScript	HTML5 semantic elements, CSS3 box model, DOM manipulation, event handling, arrow functions, closures, promises and async/await, event loops, AJAX Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
2	React.js Frontend Development	component lifecycle, state and props, hooks, virtual DOM, React Router, conditional rendering, forms and validation, context API Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
3	Node.js and Express Backend	Node.js runtime, REPL, CommonJS modules, express router, middleware stack, REST API, error handling, JWT authentication Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
4	Database Integration	MongoDB schema design, CRUD operations, aggregation pipeline, Mongoose ORM, SQL vs NoSQL, indexing strategies, transactions Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
5	MVC and Design Patterns	model-view-controller, model-view, separation of concerns, repository pattern, service layer, dependency injection Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1
6	Security and Deployment	CSP headers, XSS prevention, CORS policy, HTTPS and TLS, OAuth2 flow, Docker containerization, CI/CD pipeline basics Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1

Total Hours: 48

Note: No questions from Self-Learning topics.

Text Books:

1. Web Development by J. Smith, Pearson, 2nd edition.

References:

1. MDN Web Docs. https://developer.mozilla.org

Assessment:

IAT-I + IAT-II: 40 Marks, End Sem: 60 Marks

Course Code | Course Name | Teaching Scheme

2345114L Web Technology Lab

Theory Practical Tutorial Credits

3 0 0 3

Lab Objectives:

1. Students will implement programs.
2. Students will write test cases.

Course Outcomes:

CO1: Demonstrate knowledge of fundamentals.

CO2: Analyze and design solutions.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Data Structures, Mathematics, Programming	00	
1	Web Development Lab	implement REST endpoints, build React components, connect MongoDB, deploy on cloud		
		Self-Learning Topics: Explore online resources for deeper understanding.	06	CO1

Total Hours: 48

Note: No questions from Self-Learning topics.

Text Books:

1. Web Development by J. Smith, Pearson, 2nd edition.

References:

1. MDN Web Docs. https://developer.mozilla.org

Assessment:

IAT-I + IAT-II: 40 Marks, End Sem: 60 Marks